How Many Parameters Can Be Special?

Uniformity in the Legendre–Lattès family

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The motivating parameter-space problem

The Legendre family is

$$E_t : \quad y^2 = x(x-1)(x-t), \qquad t \in S := \mathbb{P}^1 \setminus \{0, 1, \infty\}.$$

Multiplication by 2 descends through $x : E_t \to \mathbb{P}^1$:

$$F_t \circ x = x \circ [2], \qquad \deg F_t = 4.$$

For a constant marking c, choose

$$P_c(t) = \left(c, \sqrt{c(c-1)(c-t)}\right) \in E_t.$$

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The set we count

For distinct admissible a, b, put

$$\mathrm{CT}(a, b) = \left\{ t : \ P_a(t), P_b(t) \in E_t(\overline{\mathbb{Q}})_{\mathrm{tor}} \right\}.$$

Equivalently,

$$t \in \mathrm{CT}(a, b) \iff a, b \in \mathrm{PrePer}(F_t).$$

$\#\,\mathrm{CT}(a, b) \leq M$ independently of a, b?

From qualitative finiteness to uniformity

What was known

- Masser–Zannier: simultaneous torsion is finite for each fixed distinct pair.
- DeMarco–Wang–Ye: a dynamical proof using bifurcation measures for the Legendre–Lattès family.
- Fu: effective parameter-space uniform bound for unicritical polynomials.

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Two routes toward a uniform bound

Analytic/adelic route

many common parameters $\Longrightarrow$ small energy pairing,

followed by a uniform positive lower bound.

Geometric route

large fiber powers $\Longrightarrow$ proper torsion closure,

using relative Bogomolov and some rigidity lemmas.

The second route is inspired by DeMarco–Mavraki–Ye's bounded-geometry strategy.

The constant-marking answer: the generic bound is 2

Let

$$V_{\mathrm{adm}} = \left((\mathbb{P}^1 \setminus \{0,1,\infty\})^2\right) \setminus \Delta_{a=b}.$$

Sharp generic theorem

There is a proper Zariski-closed subset

$$\mathcal{E}_3 \subsetneq V_{\mathrm{adm}}$$

such that

$$\boxed{\#\,\mathrm{CT}(a,b) \le 2} \qquad \text{for } (a,b) \in V_{\mathrm{adm}} \setminus \mathcal{E}_3.$$

Moreover,

$$\{(a,b) : \#\,\mathrm{CT}(a,b) \ge 2\}$$

is Zariski dense in V_{adm}. Thus 2 is optimal.

A broader result: uniform bounded geometry

Fix an elliptic scheme

$$\pi : \mathcal{E} \longrightarrow S.$$

Let $\mathbf{P} = (P_1, \dots, P_k)$ be sections of degree at most d, and let $X \subseteq S$ have degree at most D. Define

$$Z_{\mathbf{P},X} = \overline{\left\{x \in X : \ P_1(x), \dots, P_k(x) \in \mathcal{E}_x(\overline{\mathbb{Q}})_{\mathrm{tor}}\right\}}^{\mathrm{Zar}}.$$

Bounded-geometry theorem

There are constants

$$N = N(\mathcal{E}/S, k, d, D), \qquad B = B(\mathcal{E}/S, k, d, D)$$

such that $Z_{\mathbf{P},X}$ has at most N irreducible components, each of degree at most B.

What this gives for Lattès families

Let

$$\kappa : \mathcal{E} \longrightarrow \mathcal{P} = \mathcal{E}/\{\pm 1\}, \qquad f_n \circ \kappa = \kappa \circ [n].$$

For bounded-degree marked sections $a, b : S \to \mathcal{P}$, bounded geometry gives the alternative

$$\boxed{\begin{array}{c} [u]P_a + [v]P_b = T \text{ after a finite cover} \\ \text{or} \\ \#\{t : \ a(t), b(t) \in \mathrm{PrePer}(f_{n,t})\} \leq M. \end{array}}$$

Here $(u, v) \neq (0, 0)$ and T is torsion.